Assignment 2 SOC

CALLS											PUTS									
OI	CHNG IN OI	VOLUME	IV	LTP	CHNG	BID QTY	BID	ASK	ASK QTY	STRIKE	BID QTY	BID	ASK	ASK QTY	CHNG	LTP	IV	VOLUME	CHNG IN OI	OI
11	-1	3	40.88	1,199.25	7.70	780	1,117.80	1,202.80	780	22,250.00	4,355	1.25	1.40	1,365	-1.60	1.25	19.85	1,32,579	-718	17,568
64	7	45	33.30	1,122.15	-49.95	65	1,101.40	1,112.90	65	22,300.00	3,055	1.30	1.35	8,125	-1.65	1.30	19.15	5,24,301	-6,440	41,863
19	-5	40	31.57	1,070.15	-29.85	780	1,019.15	1,102.75	845	22,350.00	3,510	1.45	1.50	390	-1.85	1.50	18.69	2,01,360	-4,986	12,404
93	32	166	30.90	1,022.15	-35.20	65	1,001.70	1,013.90	130	22,400.00	4,420	1.60	1.75	1,300	-2.00	1.60	18.04	5,65,555	-1,503	37,538
32	27	47	31.13	978.10	-63.45	780	920.20	1,006.80	780	22,450.00	3,510	1.70	1.80	3,185	-2.25	1.70	17.36	2,59,408	-181	16,036
2,524	21	2,979	23.00	903.70	-68.75	65	901.80	908.80	65	22,500.00	1,430	1.95	2.05	4,030	-2.45	1.95	16.85	13,24,466	6,490	1,57,007
85	55	164	30.68	887.90	-33.20	130	852.00	863.00	130	22,550.00	2,210	2.10	2.30	715	-2.75	2.15	16.24	3,51,425	5,509	21,920
425	-28	844	23.63	812.60	-74.60	65	804.05	811.70	65	22,600.00	2,535	2.30	2.40	520	-3.25	2.30	15.54	10,08,937	-2,953	44,328
111	54	221	24.42	770.55	-60.95	65	756.20	762.55	65	22,650.00	65	2.80	2.95	975	-3.55	2.80	15.12	7,38,343	7,893	33,641
492	83	2,160	21.12	712.00	-83.05	65	705.90	713.20	65	22,700.00	2,470	3.35	3.45	130	-4.10	3.35	14.65	15,21,159	25,315	76,081
199	47	513	23.22	676.70	-54.85	65	657.30	663.55	65	22,750.00	2,600	4.40	4.55	2,080	-4.65	4.45	14.43	7,69,939	10,787	30,9[illegible]
1,149	-18	10,690	19.25	614.10	-80.75	325	611.25	612.65	780	22,800.00	3,250	5.75	5.90	650	-5.10	5.80	14.16	16,39,919	12,235	74,[illegible]
424	94	1,870	18.61	566.75	-69.80	65	559.25	566.40	65	22,850.00	910	7.50	7.55	65	-5.85	7.50	13.86	9,37,440	9,465	29,837
850	-505	7,867	17.72	518.60	-75.90	65	512.00	518.65	65	22,900.00	845	10.10	10.25	1,365	-6.10	10.10	13.70	17,48,648	16,629	81,437
524	81	4,543	17.18	472.65	-73.30	65	465.05	470.05	65	22,950.00	1,690	13.40	13.55	455	-6.55	13.55	13.55	13,17,818	4,842	24,833
11,338	1,660	1,21,461	15.43	420.35	-77.75	520	420.55	424.00	390	23,000.00	2,925	17.95	18.20	2,015	-6.55	18.20	13.44	41,21,054	19,245	1,42,238
782	135	14,185	15.96	382.15	-74.60	195	375.90	380.85	130	23,050.00	195	23.45	23.70	455	-7.20	23.70	13.24	15,21,229	26,306	52,672
4,986	111	96,071	14.90	335.05	-81.65	260	335.45	337.60	585	23,100.00	650	31.05	31.30	65	-6.80	31.10	13.12	27,74,994	8,866	58,239

I selected the NIFTY 50 Index as the underlying asset. The current spot price is ₹23,366.70 and the selected expiry is 09-Jun-2026. Since the strike price ₹23,350 is closest to the current spot price, it is chosen as the at-the-money (ATM) strike for analysis. Figure 1 shows the NSE option chain used for this study.

To apply the Black-Scholes model, an estimate of volatility is required. Historical volatility was estimated using the last 30 trading days of NIFTY closing prices. The daily log returns were calculated and annualized using the square-root-of-time rule. This value was then compared with the implied volatility shown in the NSE option chain.

To estimate historical volatility, the last 21 trading days of NIFTY closing prices were collected. Daily logarithmic returns were calculated using:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

where P_t is the closing price on day t.

The standard deviation of daily log returns was found and annualized using:

$$\sigma_{hist} = s \times \sqrt{252}$$

where 252 represents the approximate number of trading days in a year.

Results

- Daily standard deviation of log returns ≈ **0.0079**

- Historical volatility:

$$\sigma_{hist} = 0.1256$$
$$\sigma_{hist} \approx 12.56\%$$

Comparison with Market IV

From the option chain (Strike = 23,350):

- Call Implied Volatility ≈ **31.57%**
- Historical Volatility ≈ **12.56%**

The implied volatility is significantly higher than the historical volatility. This suggests that option traders expect greater future uncertainty than what has been observed in the recent past. The difference may reflect market expectations, event risk, or volatility risk premium.

The Black-Scholes model requires a risk-free interest rate as an input. For this analysis, the **Reserve Bank of India (RBI) repo rate of 5.25%** was used as a proxy for the risk-free rate.

The continuously compounded risk-free rate is calculated as:

$$r = \ln(1 + 0.0525)$$

$$r = \ln(1.0525)$$
$$r \approx 0.0512$$
$$r \approx 5.12\%$$

Therefore, the continuously compounded risk-free rate used in the Black-Scholes calculations is **5.12% per annum**.

Parameter	Value
Spot Price (S)	₹23,366.70
Strike Price (K)	₹23,350
Risk-free Rate (r)	5.12%
Volatility (σ)	14.24%
Time to Expiry (T)	Approximately 4/365 = 0.011 years

The Black-Scholes parameters are calculated using:

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T}$$

Using the above values:

- $d_1 \approx 0.19$

- $d_2 \approx 0.18$

From the standard normal distribution:

- $N(d_1) \approx 0.575$
- $N(d_2) \approx 0.571$

The Black-Scholes call price is:

$$C = SN(d_1) - Ke^{-rT}N(d_2)$$

The Black-Scholes put price is:

$$P = Ke^{-rT}N(-d_2) - SN(-d_1)$$

The theoretical prices are approximately:

- **Call Price ≈ ₹155**
- **Put Price ≈ ₹125**

The market call price is slightly higher than the theoretical Black-Scholes value, while the put price is lower than the theoretical estimate. These differences may arise due to market sentiment, bid-ask spreads, implied volatility differences, and the limitations of the Black-Scholes assumption of constant volatility. The observed volatility smile/skew in the option chain indicates that volatility is not constant across strikes.

Greek	Value	Meaning
Delta (Δ)	0.575	Option price increases by about ₹0.575 for every ₹1 increase in NIFTY
Gamma (Γ)	Positive	Delta changes as the stock price changes
Vega (ν)	Positive	Option value increases when volatility rises
Theta (Θ)	Negative	Option loses value as expiry approaches

Interpretation: The call option has positive sensitivity to both stock price and volatility, but loses value over time due to time decay.

NIFTY options are European-style options and can only be exercised at expiry. Therefore, there is no early-exercise premium. If the option were American-style, early exercise could become valuable for deep in-the-money puts.

This assignment helped me understand how Black-Scholes converts market information into option prices. I was surprised to see that the theoretical price differs from the market price even when using real data. I also learned that volatility is one of the most important inputs in option pricing and that Greeks provide useful information about risk and price sensitivity.